RAMAIAH Institute of Technology

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Information Science and Engineering

Question for the Laboratory: Semester End Examination (SEE)

Course Name: Advanced Java Lab	Term: 09th February 2026 to June 6th, 2026
Course Code: ISL48	CIE Marks: 50
Credits: 0:0:1	Exam Hours: 03, Max. Marks: 50

<table>
<tr><td>1.</td><td>Design a class called Complex with two data members, real and imaginary. Perform the following operations. Use default and parameterised constructors to initialise the values of data members. Use display() to display a Complex number in proper format. Use multiply() to multiply two Complex Numbers and return the resultant Complex Number. Use MainClass to create complex objects and perform computation.</td></tr>
<tr><td>2.</td><td>Create a Bank class with account number, customer name, customer address, balance, email and phone number. Perform the following operations
• default and parameterised constructors to initialise the values of data members,
• display() to display the details of the Bank object.
• deposit() to perform a deposit operation
• withdraw() to perform the withdraw operation
• getBalance() to view the current balance of the account. Note: Ensure that the minimum balance is Rs.500/-</td></tr>
<tr><td>3.</td><td>Write a Java Program for the following Payroll System. Create a package called PayrollManagement. Define an interface Payable with a method calculate(). Create a super class called Employee with the following data members name, age, salary and designation. Use a parameterised constructor to initialize all the data members. The Employee class is inherited by three sub-classes namely FullTime, PartTime and Intern and these subclasses implement the Payable interface. Full-Time Employee receives monthly salary and Part-Time Employee receives hourly wages. Intern takes a fixed stipend. Create another package “Company” and import these classes to complete the payroll system. Create a reference to the Employee class and identify the types of employee , compute salary and display.</td></tr>
<tr><td>4.</td><td>Create a Queue class and implement enqueue, dequeue and display operations. Create Custom Exceptions to handle Queue Overflow and Queue Underflow. Write a menu driven program to perform the queue operations.</td></tr>
<tr><td>5.</td><td>Define a package named stringoperations to encapsulate the string processing functionality. Create an interface named StringManipulator for string operations:
• String reverse(String input): Reverses the given string.
• String toUpperCase(String input): Converts the string to uppercase.
• String toLowerCase(String input): Converts the string to lowercase.
• String concatenate(String str1, String str2): Concatenates two strings.
• int countVowels(String input): Counts the number of vowels in the string.
• int wordCount(String input): Counts the number of words in the string.
Implement the interface in a class named StringProcessor within the stringoperations package, providing concrete implementations for all methods. Create a main class in a separate package application, where an object of StringProcessor is created. The user is prompted to enter a string. Each method of the interface is tested, and its output is displayed.</td></tr>
<tr><td>6.</td><td>Create a Palindrome Checker program using StringBuffer that verifies if a given string is a palindrome. The program should implement user-defined exception handling for the following: If the input string contains non-alphabetical characters, throw a custom exception InvalidInputException. If the string length is less than 3 characters, throw a custom exception ShortStringException. If the string</td></tr>
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	is a palindrome (irrespective of the case), print a message indicating that it is a palindrome. If the string is not a palindrome, print a message indicating that it is not a palindrome.
7.	Create a Password Security Application in Java that takes a user's password as input and performs the following operations. • Check if the password contains at least one uppercase letter, one lowercase letter, and one digit. • Count the number of special characters in the password. • Mask the password by replacing all characters with * (except the first and last character). • Reverse the password and display it (for security encryption demonstration). • Append a random security token (e.g., "@123!") to the password. • Replace all vowels (a, e, i, o, u) with # to make it difficult to read.
8.	Write a Program that simulates a telephone that records missed incoming calls. For each missed call, telephone number of origins, and name of the caller if the name is available. For unlisted numbers, set the name to "private caller". Choose or extend the most appropriate collection class and provide the following features. Store upto 3 numbers. When the fourth call comes in, it is stored and the oldest call is deleted so that no more than 3 numbers are ever recorded. Iterate through the missed call collection and identify the calls to delete and remove them from the collection . Display the missed caller details. Define a class that enables all of the above mentioned operations.
9.	Write a Java program using user-defined storage classes to create a book database and store it in a Collection List. Books collection should include title, author, publisher and price. Write a method to sort the books in ascending order of price and store it in another List. Maintain the book details with respect to a unique book id. Prompt for an author name and list all the books with the same author name. Create a new list holding all the book details with price greater than a user specified price.
10.	Write a program to create generic Stack class with push(), pop(), clear(), isEmpty() and display() methods. Demonstrate creating Stack of String and Integer objects. Write a menu driven program to show the working of the Stack data structure.
11.	Create a desktop java application using swings to enable a user to enter student information such as name, usn, age, address, cgpa, category. Perform validations on all the fields. Display appropriate messages in pop up boxes to indicate wrong entries. Include a button insert which on clicking enters the GUI data into a collection and display the collection on the console.
12.	Write a java program using Swing to validate user login information using dialog boxes. Once validated, allow the user to enter the customer id, if the person is a new customer, else check whether the customer exists in a collection and obtain the customer id. Allow the user to enter the item purchased by giving the item id and quantity purchased. On clicking of a button, the item name and the total cost in a message dialog box.

Evaluation Criteria (Max. Marks:50)	
Write Up	08 Marks
Conduction and Execution	35 Marks
Viva	07 Marks
Change of Program	Write up is evaluated for only 3 marks

Course Co-ordinator **HoD**